Chapter 1
History of Computing

Computing can be defined as any goal-oriented activity requiring, benefiting from, or creating computers. Thus, computing includes designing and building hardware and software systems for a wide range of purposes; processing, structuring, and managing various kinds of information; doing scientific studies using computers; making computer systems behave intelligently; creating and using communications and entertainment media; finding and gathering information relevant to any particular purpose, and so on. The history of computing is longer than the history of computing hardware and modern computing technology.

The desire to improve reasoning and mathematical accuracy led ancient civilizations of people to develop formal logic, formulas, tables and patterns. Algebra, logarithms, and analytical geometry are being developed to assist us in complex mathematics.

1.1 Prehistory of computing

Computing is intimately tied to the representation of numbers and thus counting. The first method of counting was counting on fingers. This evolved into sign language for the hand-to-eye communication of numbers which, while not writing, gave way to written numbers.

Tallies made by carving notches in wood, bone, and stone were used for at least forty thousand years. Stone Age cultures, including ancient Native American groups, used tallies for gambling with horses, slaves, personal services and trade-goods.

Roman numerals evolved from this primitive system of cutting notches. We will briefly discuss some of the path-breaking inventions in the field of computing devices.

1. ABACUS

It took over generations for early man to build mechanical devices for counting large numbers. The first calculating device called ABACUS was developed by the Egyptian and Chinese people. The word ABACUS means calculating board. It consisted of sticks in horizontal positions on which were inserted sets of pebbles. It has a number of horizontal bars each having ten beads. Horizontal bars represent units, tens, hundreds, etc.

2. Napier's bones

English mathematician John Napier built a mechanical device for the purpose of multiplication in 1617 A D. The device was known as Napier's bones.

3. Slide Rule

English mathematician Edmund Gunter developed the slide rule. This machine could perform operations like addition, subtraction, multiplication, and division. It was widely used in Europe in 16th century.

4. Pascal's Adding and Subtractin Machine

Pascal developed a machine at the age of 19 that could add and subtract. The machine consisted of wheels, gears and cylinders.

5. Leibniz's Multiplication and Dividing Machine

The German philosopher and mathematician Gottfried Leibniz built around 1673 a mechanical device that could both multiply and divide.

6. Babbage's Analytical Engine

It was in the year 1823 that a famous English man Charles Babbage built a mechanical machine to do complex mathematical calculations. It was called *difference engine*. Later he developed a general-purpose calculating machine called *analytical engine*. You should know that Charles Babbage is called the *father of computer*.

7. Mechanical and Electrical Calculator

In the beginning of 19th century the mechanical calculator was developed to perform all sorts of mathematical calculations. Up to the 1960s it was widely used.

1.2 History of computer hardware

We all use personal computers and we all take them for granted in our everyday lives. It's easy to forget that PCs have only been around for a couple of decades, and initially were nowhere near the powerhouses we have on our desks today.

For example, did you know that the first "portable" computer weighed 25 kg (55 lb) and cost close to $20,000, that the first laser printer was big enough to fill up most of a room, or that you basically had to build the first Apple computer yourself?

THE FIRST COMPUTER MOUSE

The first computer mouse was invented in 1963 by Douglas Engelbart at the Stanford Research Institute. (He is also one of the inventors of hypertext.) The first mouse used two wheels positioned at a 90-degree angle to each other to keep track of the movement. The ball mouse wasn't invented until 1972, and the optical mouse was invented circa 1980 although it didn't come to popular use until much later.
Douglas Engelbart never received any royalties for his invention and his patent had run out by the time the mouse became commonplace in the era of home PCs.

THE FIRST PORTABLE COMPUTER

Well, perhaps that should be "movable" computer. The IBM 5100 Portable Computer was introduced in 1975, weighed 25 kg (55 lb), was the size of a small suitcase and needed external power to operate. It held everything in the same unit, packing in a processor, ROM (several hundreds of KB) and RAM (16-64 KB), a five-inch CRT display, keyboard and a tape drive, which was an amazing feat at the time. It also came with built-in BASIC and/or APL. The different models of the IBM 5100 sold for $8,975 – $19,975.

THE FIRST LAPTOP COMPUTER

The first laptop computer (or notebook) was the Grid Compass 1100 (called the GRiD) and was designed in 1979 by a British industrial designer, Bill Moggridge. The computer didn't start selling until 1982, then featuring a 320×200 screen, an Intel 8086 processor, 340 KB of magnetic bubble memory (a now obsolete, non-volatile memory type) and a 1200 bps modem. It weighed 5 kg (11 lb) and cost $8-10,000. The GRiD was mainly used by NASA and the US military.

THE FIRST IBM PC

The IBM Personal Computer was introduced in 1981 as the IBM 5150. The platform became so pervasive in the 80s that although the term "personal computer" had been in use since the early 70s, a PC became synonymous with an IBM PC-compatible computer.

During its development, the IBM 5150 had been internally referred to as "Project Chess" and was created by a team of 12 people headed by Don Estridge and Larry Potter. To speed up development and cut costs, IBM had decided to use off-the-shelf parts, something that they normally wouldn't do.

The first IBM PC had an Intel 8088 processor, 64 KB of RAM (extendible to 256 KB), a floppy disk drive (which could be used to boot the computer with a rebranded version of MS-DOS (PC-DOS)) and a CGA or monochrome video card. The machine also had a version of Microsoft BASIC in ROM. On the first IBM PC the optional 10 MB hard disk drive could only be installed if the original power supply was replaced.

THE FIRST APPLE COMPUTER

The first Apple personal computers (Apple I) were designed and hand-built by Steve Wozniak. The Apple I went on sale in 1976 for the price of $666.66. Only about 200 units were produced. The Apple I was basically just a motherboard with a processor, a total of 8KB of RAM, a display interface and some additional functionality. To have a working computer, the buyer would have to add a power supply, a keyboard and a display (and a case to keep mount it all in).

THE FIRST RAM

Arguably the first (writable) random access memory was Magnetic Core Memory (also called Ferrite-Core Memory) and was invented in 1951 as a result of work done by An Wang at Harvard University's Computation Lab and Jay Forrester at MIT.

Core memory was a family of related technologies that used the magnetic properties of materials to give them a similar functionality to transistors. They stored their information using the polarity of tiny, magnetic ceramic rings with wires threaded through them. Unlike today's RAM, core memory could keep its information even after the power was turned off.

Core memory was common until it was replaced by integrated silicon RAM chips in the 1970s.

THE FIRST HARD DISK DRIVE

The IBM Model 350 Disk File was the first hard disk drive and was part of the IBM 305 RAMAC computer that IBM started delivering in 1956 (mainly intended for business accounting). It had 50 24-inch discs that together could store about 4.4 MB of data. The Model 350 spun at 1200 rpm, had a data transfer rate of 8,800 characters per second and an access time of approximately one second.

THE FIRST LASER PRINTER

The laser printer was invented by Gary Starkweather at XEROX in 1969. His initial prototype was a modified laser copier where he had disabled the imaging system and introduced a spinning drum with eight mirrored sides. The first commercial implementation of a laser printer didn't happen until IBM released the IBM model 3800 in 1976. It could pretty much fill up a room on its own.

THE FIRST WEB SERVER

And since the Web is such an integral part of today's computer experience, we couldn't help but include another first: The first web server was a NeXT workstation that Tim Berners-Lee used when he invented the World Wide Web at CERN. The first web page was put online on August 6, 1991.

The computer had a note on it that said, "This machine is a server. DO NOT POWER IT DOWN!!" Understandable, considering that if you had shut it down in the early days you would have shut down the entire WWW. :)

Hardwares: Memory

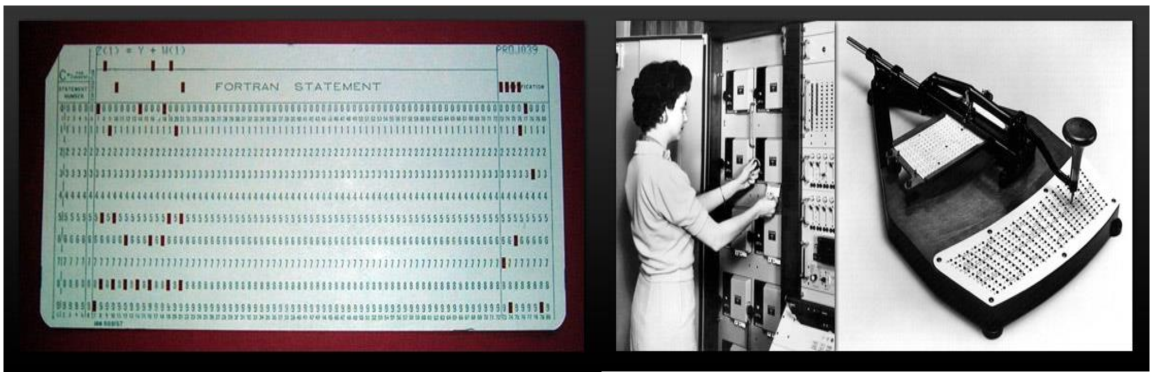
THE SELECTRON TUBE

The Selectron tube had a capacity of 256 to 4096 bits (32 to 512 bytes). The 4096-bit Selectron was 10 inches long and 3 inches wide. Originally developed in 1946, the memory storage device proved expensive and suffered from production problems, so it never became a success.

Above: *The 1024-bit Selectron.*

PUNCH CARDS

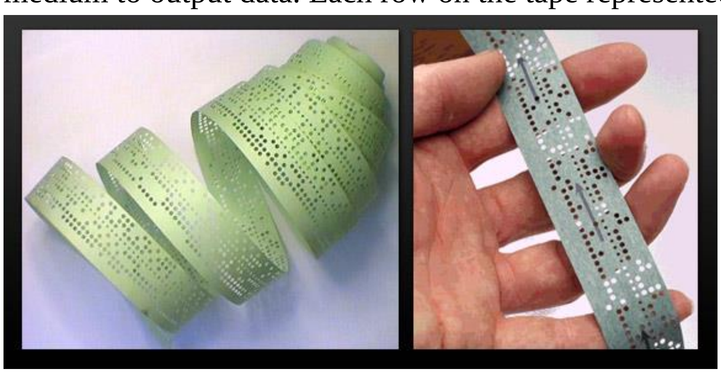
Early computers often used punch cards for input both of programs and data. Punch cards were in common use until the mid-1970s. It should be noted that the use of punch cards predates computers. They were used as early as 1725 in the textile industry (for controlling mechanized textile looms).



Above: *Card from a Fortran program: Z(1) = Y + W(1)* ***Above left:*** *Punch card reader.* ***Above right:*** *Punch card writer.*

PUNCHED TAPE

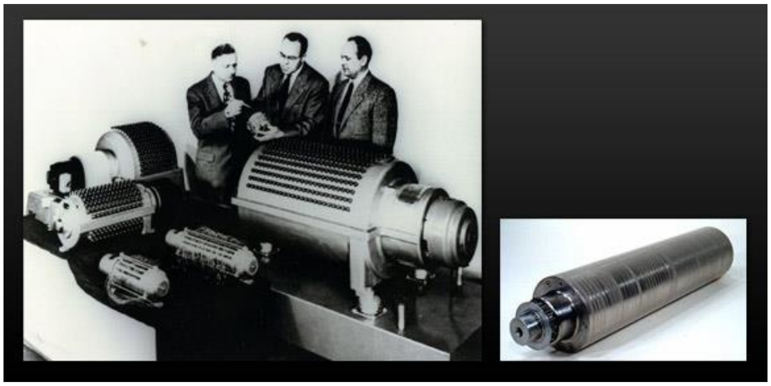
Same as with punch cards, punched tape was originally pioneered by the textile industry for use with mechanized looms. For computers, punch tape could be used for data input but also as a medium to output data. Each row on the tape represented one character.



Above: *8-level punch tape (8 holes per row).*

MAGNETIC DRUM MEMORY

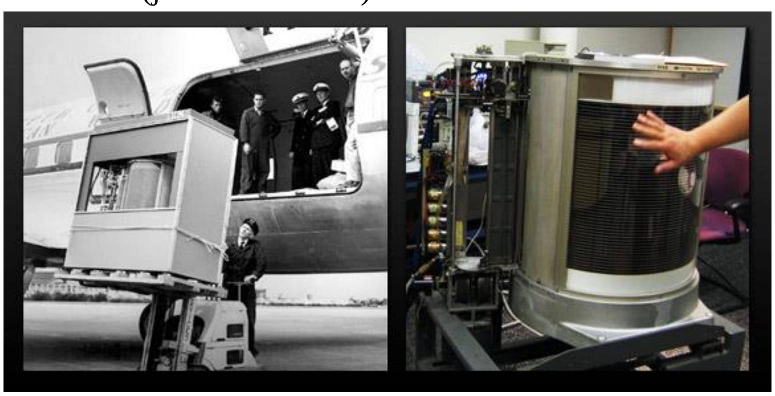
Invented all the way back in 1932 (in Austria), it was widely used in the 1950s and 60s as the main working memory of computers. In the mid-1950s, magnetic drum memory had a capacity of around 10 kB.



Above left: *The magnetic Drum Memory of the UNIVAC computer.* ***Above right:*** *A 16-inch-long drum from the IBM 650 computer. It had 40 tracks, 10 kB of storage space, and spun at 12,500 revolutions per minute.*

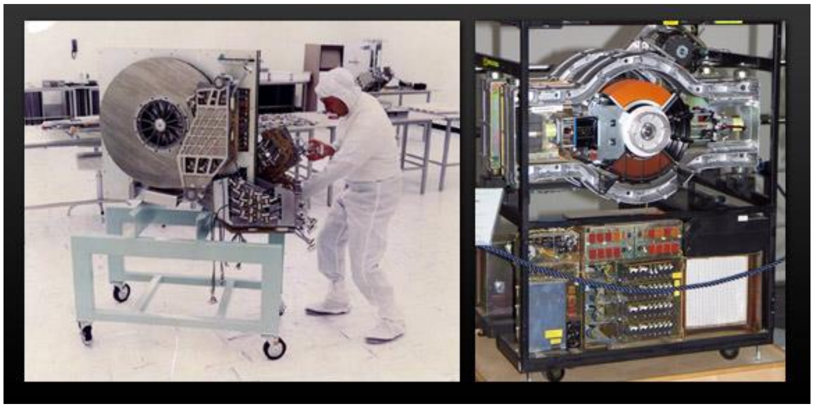
THE HARD DISK DRIVE

The first hard disk drive was the IBM Model 350 Disk File that came with the IBM 305 RAMAC computer in 1956. It had 50 24-inch discs with a total storage capacity of 5 million characters (just under 5 MB).



Above: *IBM Model 350, the first-ever hard disk drive.*

The first hard drive to have more than 1 GB in capacity was the IBM 3380 in 1980 (it could store 2.52 GB). It was the size of a refrigerator, weighed 550 pounds (250 kg), and the price when it was introduced ranged from $81,000 to $142,400.



Above left: *A 250 MB hard disk drive from 1979.* ***Above right:*** *The IBM 3380 from 1980, the first gigabyte-capacity hard disk drive.*

THE LASERDISC

We mention it here mainly because it was the precursor to the CD-ROM and other optical storage solutions. It was mainly used for movies. The first commercially available laserdisc system was available on the market late in 1978 (then called Laser Videodisc and the more funkily branded DiscoVision) and were 11.81 inches (30 cm) in diameter. The discs could have up to 60 minutes of audio/video on each side. The first laserdiscs had entirely analog content. The basic technology behind laserdiscs was invented all the way back in 1958.

Above left: *A Laserdisc next to a regular DVD.* ***Above right:*** *Another Laserdisc.*

THE FLOPPY DISC

The diskette, or floppy disk (named so because they were flexible), was invented by IBM and in common use from the mid-1970s to the late 1990s. The first floppy disks were 8 inches, and later in came 5.25 and 3.5-inch formats. The first floppy disk, introduced in 1971, had a capacity of 79.7 kB, and was read-only. A read-write version came a year later.

Above left: *An 8-inch floppy and floppy drive next to a regular 3.5-inch floppy disk.* ***Above right:*** *The convenience of easily removable storage media.*

MAGNETIC TAPE

Magnetic tape was first used for data storage in 1951. The tape device was called UNISERVO and was the main I/O device on the UNIVAC I computer. The effective transfer rate for the UNISERVO was about 7,200 characters per second. The tapes were metal and 1200 feet long (365 meters) and therefore very heavy.

Above left: *The row of tape drives for the UNIVAC I computer.* ***Above right:*** *The IBM 3410 Magnetic Tape Subsystem, introduced in 1971.*

And of course, we can’t mention magnetic tape without also mentioning the standard compact cassette, which was a popular way of data storage for personal computers in the late 70s and 80s. Typical data rates for compact cassettes were 2,000 bit/s. You could store about 660 kB per side on a 90-minute tape.

Above left: *The standard compact cassette.* ***Above right:*** *The Commodore Datassette is sure to bring up fond memories for people who grew up in the 80s.*

1.3 History of software

Software can be defined as programmed instructions stored in the memory of stored-program digital computers for execution by the processor. The design for what would have been the first piece of software was written by Ada Lovelace in the 19th century but was never implemented.

Alan Turing is credited with being the first person to come up with a theory for software, which led to the two academic fields of computer science and software engineering.

1945

Konrad Zuse

- Konrad Zuse began work on Plankalkul (Plan Calculus), the first algorithmic programming language, with an aim of creating the theoretical preconditions for the formulation of problems of a general nature. Seven years earlier, Zuse had developed and built the world´s first binary digital computer, the Z1. He completed the first fully functional program-controlled electromechanical digital computer, the Z3, in 1941. Only the Z4 — the most sophisticated of his creations — survived World War II.

1948

Claude Shannon

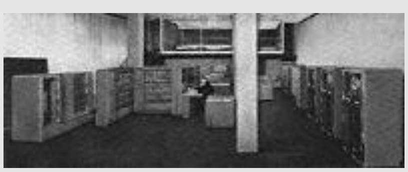
- Claude Shannon´s "The Mathematical Theory of Communication" showed engineers how to code data so they could check for accuracy after transmission between computers. Shannon identified the bit as the fundamental unit of data and, coincidentally, the basic unit of computation.

1952

Grace Hopper with UNIVAC 1

- Grace Hopper completes the A-0 Compiler. In 1952, mathematician Grace Hopper completed what is considered to be the first compiler, a program that allows a computer user to use English-like words instead of numbers. Other compilers based on A-0 followed: ARITH-MATIC, MATH-MATIC and FLOW-MATIC [software]

1953



IBM 701

- John Backus completed speedcoding for IBM´s 701 computer. Although speedcoding demanded more memory and compute time, it trimmed weeks off of the programming schedule.

1955

- Herbert Simon and Allen Newell unveiled Logic Theorist software that supplied rules of reasoning and proved symbolic logic theorems. The release of Logic Theorist marked a milestone in establishing the field of artificial intelligence.

1956

- In the mid-fifties resources for scientific and engineering computing were in short supply and were very precious. The first operating system for the IBM 704 reflected the cooperation of Bob Patrick of General Motors Research and Owen Mock of North American Aviation. Called the GM-NAA I/O System, it provided batch processing and increased the number of completed jobs per shift with no increase in cost. Some version of the system was used in about forty 704 installations.

MIT Whirlwind

- At MIT, researchers began experimentation on direct keyboard input on computers, a precursor to today´s normal mode of operation. Doug Ross wrote a memo advocating direct access in February; five months later, the Whirlwind aided in such an experiment.

1957

UNIVAC MATH-MATIC

- Sperry Rand released a commercial compiler for its UNIVAC. Developed by Grace Hopper as a refinement of her earlier innovation, the A-0 compiler, the new version was called MATH-MATIC. Earlier work on the A-0 and A-2 compilers led to the development of the first English-language business data processing compiler, B-0 (FLOW-MATIC), also completed in 1957. FLOW-MATIC served as a model on which to build with input from other sources.


Programmer's
Reference Manual

THE FORTRAN
AUTOMATIC CODING SYSTEM
FOR THE
IBM 704 EDPM

IBM 704 FORTRAN

- A new language, FORTRAN (short for FORmula TRANslator), enabled a computer to perform a repetitive task from a single set of instructions by using loops. The first commercial FORTRAN program ran at Westinghouse, producing a missing comma diagnostic. A successful attempt followed.

1959

ERMA characters

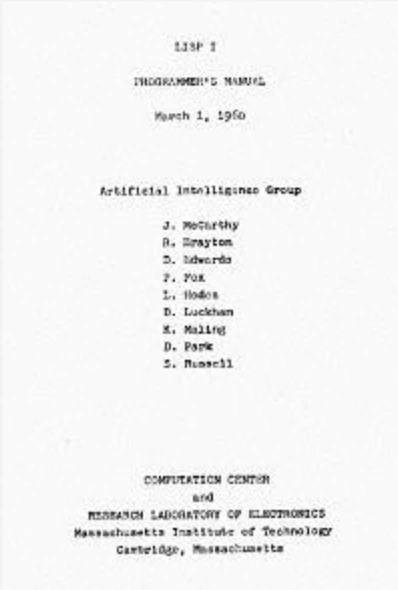
- ERMA, the Electronic Recording Method of Accounting, digitized checking for the Bank of America by creating a computer-readable font. A special scanner read account numbers preprinted on checks in magnetic ink.

1960

COBOL design team

- A team drawn from several computer manufacturers and the Pentagon developed COBOL, Common Business Oriented Language. Designed for business use, early COBOL efforts aimed for easy readability of computer programs and as much machine independence as possible. Designers hoped a COBOL program would run on any computer for which a compiler existed with only minimal modifications.

  Howard Bromberg, an impatient member of the committee in charge of creating COBOL, had this tombstone made out of fear that the language had no future. However, COBOL has survived to this day.



LISP I

PROGRAMMER'S MANUAL

March 1, 1960

Artificial Intelligence Group

J. McCarthy
R. Brayton
D. Edwards
P. Fox
L. Hodes
D. Luckham
K. Maling
D. Park
S. Russell

COMPUTATION CENTER
and
RESEARCH LABORATORY OF ELECTRONICS
Massachusetts Institute of Technology
Cambridge, Massachusetts

LISP Programmer´s Reference

- LISP made its debut as the first computer language designed for writing artificial intelligence programs. Created by John McCarthy, LISP offered programmers flexibility in organization.

C.A.R. Hoare circa 1965

- Quicksort is developed. Working for the British computer company Elliott Brothers, C. A. R. Hoare developed Quicksort, an algorithm that would go on to become the most used sorting method in the world. Quicksort used a series of elements called “pivots” that allowed for fast sorting. C.A.R. Hoare was knighted by Queen Elizabeth II in 2000.

1962

Spacewar! on PDP-1

- MIT students Slug Russell, Shag Graetz, and Alan Kotok wrote SpaceWar!, considered the first interactive computer game. First played at MIT on DEC´s PDP-1, the large-scope display featured interactive, shoot´em-up graphics that inspired future video games. Dueling players fired at each other´s spaceships and used early versions of joysticks to manipulate away from the central gravitational force of a sun as well as from the enemy ship.

Ken Iverson lecturing at blackboard.

- Published in 1962, Kenneth Iverson's book *A Programming Language* detailed a form of mathematical notation that he had developed in the late 1950s while an assistant professor at Harvard University. IBM hired Iverson and it was there that APL evolved into a practical programming language. APL was widely used in scientific, financial, and especially actuarial applications. Powerful functions and operators in APL are expressed with special characters, resulting in very concise programs.

1963

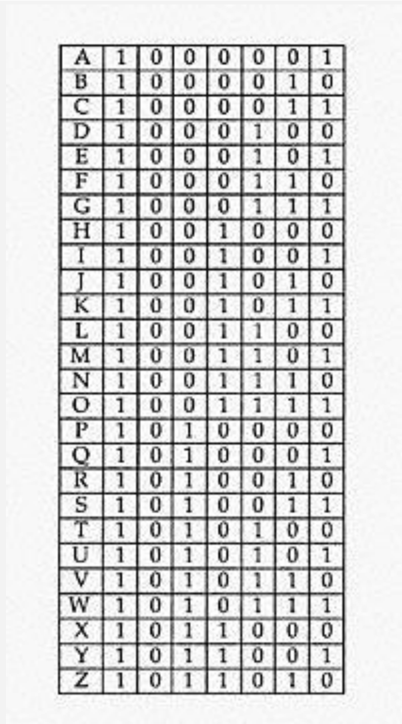
Unclassified

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
LINCOLN LABORATORY

SKETCHPAD: A MAN-MACHINE GRAPHICAL
COMMUNICATION SYSTEM

Sketchpad document

- Ivan Sutherland published Sketchpad, an interactive, real time computer drawing system, as his MIT doctoral thesis. Using a light pen and Sketchpad, a designer could draw and manipulate geometric figures on the screen.

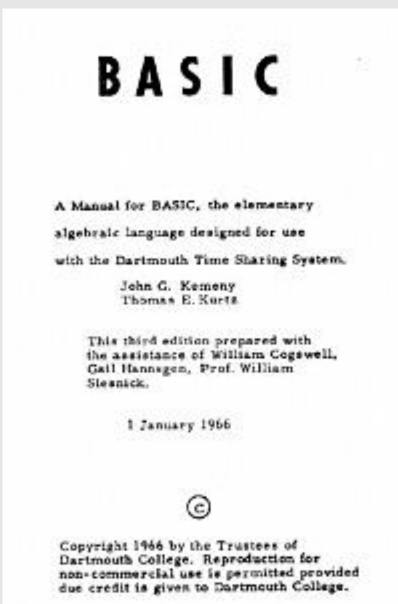


A	1	0	0	0	0	0	1
B	1	0	0	0	0	1	0
C	1	0	0	0	0	1	1
D	1	0	0	0	1	0	0
E	1	0	0	0	1	0	1
F	1	0	0	0	1	1	0
G	1	0	0	0	1	1	1
H	1	0	0	1	0	0	0
I	1	0	0	1	0	0	1
J	1	0	0	1	0	1	0
K	1	0	0	1	0	1	1
L	1	0	0	1	1	0	0
M	1	0	0	1	1	0	1
N	1	0	0	1	1	1	0
O	1	0	0	1	1	1	1
P	1	0	1	0	0	0	0
Q	1	0	1	0	0	0	1
R	1	0	1	0	0	1	0
S	1	0	1	0	0	1	1
T	1	0	1	0	1	0	0
U	1	0	1	0	1	0	1
V	1	0	1	0	1	1	0
W	1	0	1	0	1	1	1
X	1	0	1	1	0	0	0
Y	1	0	1	1	0	0	1
Z	1	0	1	1	0	1	0

ASCII code

- ASCII — American Standard Code for Information Interchange — permitted machines from different manufacturers to exchange data. ASCII consists of 128 unique strings of ones and zeros. Each sequence represents a letter of the English alphabet, an Arabic numeral, an assortment of punctuation marks and symbols, or a function such as a carriage return.

1964



BASIC

A Manual for BASIC, the elementary algebraic language designed for use with the Dartmouth Time Sharing System.

John G. Kemeny
Thomas E. Kurtz

This third edition prepared with the assistance of William Cogswell, Gail Hannsgen, Prof. William Slesnick.

1 January 1966

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BASIC manual

- Thomas Kurtz and John Kemeny created BASIC, an easy-to-learn programming language, for their students at Dartmouth College.

1965

- Object-oriented languages got an early boost with Simula, written by Kristen Nygaard and Ole-John Dahl. Simula grouped data and instructions into blocks called objects, each representing one facet of a system intended for simulation.

1967

Seymour Papert

- Seymour Papert designed LOGO as a computer language for children. Initially a drawing program, LOGO controlled the actions of a mechanical "turtle," which traced its path with pen on paper. Electronic turtles made their designs on a video display monitor.

  Papert emphasized creative exploration over memorization of facts: *"People give lip service to learning to learn, but if you look at curriculum in schools, most of it is about dates, fractions, and science facts; very little of it is about learning. I like to think of learning as an expertise that every one of us can acquire."*

1968

- Edsger Dijkstra´s "GO TO considered harmful" letter, published in Communications of the ACM, fired the first salvo in the structured programming wars. The ACM considered the resulting acrimony sufficiently harmful that it established a policy of no longer printing articles taking such an assertive position against a coding practice.

1969

- The RS-232-C standard for communication permitted computers and peripheral devices to transmit information serially — that is, one bit at a time. The RS-232-C protocol spelled out a purpose for a serial plug´s 25 connector pins.

UNIX "license plate"

- AT&T Bell Laboratories programmers Kenneth Thompson and Dennis Ritchie developed the UNIX operating system on a spare DEC minicomputer. UNIX combined many of the timesharing and file management features offered by Multics, from which it took its name. (Multics, a projects of the mid-1960s, represented the first effort at creating a multi-user, multi-tasking operating system.) The UNIX operating system quickly secured a wide following, particularly among engineers and scientists.

1972

- Nolan Bushnell introduced Pong and his new company, Atari video games.

1976

Command	Function
nA	append lines
±B	begin bottom of buffer
±nC	move character positions
±nD	delete characters
E	end edit and close files (normal end)
nF	find string
H	end edit, close and reopen files
I	insert characters
nJ	place strings in juxtaposition
±nK	kill lines
±nL	move down/up lines
nM	macro definition
nN	find next occurrence with autoscan
O	return to original file
±nP	move and print pages
Q	quit with no file changes
R	read library file
nS	substitute strings
±nT	type lines
± U	translate lower to upper case if U, no translation if -U
nW	write lines
nZ	sleep
±n<cr>	move and type (±nLT)

CP/MCP/M

- Gary Kildall developed CP/M, an operating system for personal computers. Widely adopted, CP/M made it possible for one version of a program to run on a variety of computers built around eight-bit microprocessors.

1977

- The U.S. government adopted IBM´s data encryption standard, the key to unlocking coded messages, to protect confidentiality within its agencies. Available to the general public as well, the standard required an eight-number key for scrambling and unscrambling data. The 70 quadrillion possible combinations made breaking the code by trial and error unlikely.

1979

Bob Frankston and Dan Brinklin

- Harvard MBA candidate Daniel Bricklin and programmer Robert Frankston developed VisiCalc, the program that made a business machine of the personal computer, for the Apple II. VisiCalc (for**Visi**ble **Calc**ulator) automated the recalculation of spreadsheets. A huge success, more than 100,000 copies sold in one year.

1981

- The MS-DOS, or Microsoft Disk Operating System, the basic software for the newly released IBM PC, established a long partnership between IBM and Microsoft, which Bill Gates and Paul Allen had founded only six years earlier.

1982

Lotus 1-2-3

- Mitch Kapor developed Lotus 1-2-3, writing the software directly into the video system of the IBM PC. By bypassing DOS, it ran much faster than its competitors. Along with the immense popularity of the IBM´s computer, Lotus owed much of its success to its working combination of spreadsheet capabilities with graphics and data retrieval capabilities.

  Kapor, who received his bachelor´s degree in an individually designed cybernetics major from Yale University in 1971, started Lotus Development Corp. to market his spreadsheet and served as its president and CEO from 1982 to 1986. He also has worked to develop policies that maximize openness and competitiveness in the computer industry.

1983

- Microsoft announced Word, originally called Multi-Tool Word, and Windows. The latter doesn´t ship until 1985, although the company said it would be on track for an April 1984 release. In a marketing blitz, Microsoft distributed 450,000 disks demonstrating its Word program in the November issue of PC World magazine.


Richard Stallman

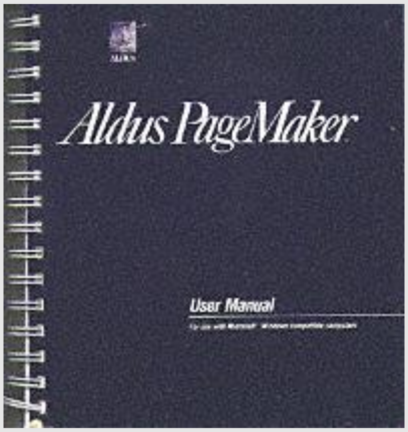
- Richard Stallman announces GNU.

  Richard Stallman, a programmer at MIT’s Artificial Intelligence Lab, experienced a significant shift in attitudes during the late 1970s. Whereas the MIT hacker culture was one of sharing and openness, the commercial software world moved towards secrecy and access to source code became ever more restricted.

  Stallman set out to develop a free alternative to the popular Unix operating system. This operating system called GNU (for *Gnu's Not Unix*) was going to be free of charge but also allow users the freedom to change and share it. Stallman founded the Free Software Foundation (FSF) based on this philosophy in 1985.

  While the GNU work did not immediately result in a full operating system, it provided the necessary tools for creating Linux. The software developed as part of the GNU project continues to form a large part of Linux, which is why the FSF asks for it to be called GNU/Linux.

1985



Aldus PageMaker

- Aldus announced its PageMaker program for use on Macintosh computers, launching an interest in desktop publishing. Two years later, Aldus released a version for IBMs and IBM-compatible computers. Developed by Paul Brainerd, who founded Aldus Corp., PageMaker allowed users to combine graphics and text easily enough to make desktop publishing practical.

  Chuck Geschke of Adobe Systems Inc., a company formed in 1994 by the merger of Adobe and Aldus, remembered: *"John Sculley, a young fellow at Apple, got three groups together — Aldus, Adobe, and Apple — and out of that came the concept of desktop publishing. Paul Brainerd of Aldus is probably the person who first uttered the phrase. All three companies then took everybody who could tie a tie and speak two sentences in a row and put them on the road, meeting with people in the printing and publishing industry and selling them on this concept. The net result was that it turned around not only the laser printer but, candidly, Apple Computer. It really turned around that whole business.*

- The C++ programming language emerged as the dominant object-oriented language in the computer industry when Bjarne Stroustrup published "The C++ Programming Language." Stroustrup, at AT&T Bell Laboratories, said his motivation stemmed from a desire to write event-driven simulations that needed a language faster than Simula. He developed a preprocessor that allowed Simula style programs to be implemented efficiently in C.

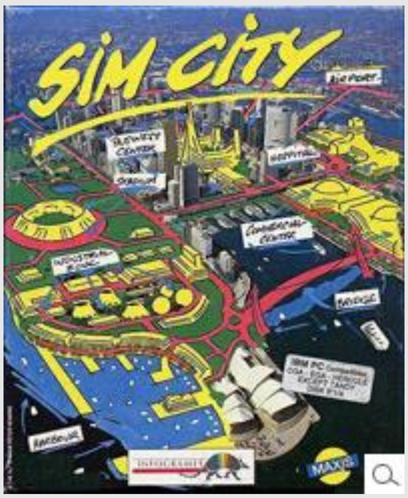
  Stroustrup wrote in the preface to "The C++ Programming Language": *"C++ is a general purpose programming language designed to make programming more enjoyable for the serious programmer. Except for minor details, C++ is a superset of the C programming language. In addition to the facilities provided by C, C++ provides flexible and efficient facilities for defining new types.... The key concept in C++ is class. A class is a user-defined type. Classes provide data hiding, guaranteed initialization of data, implicit type conversion for user-defined types, dynamic typing, user-controlled memory management, and mechanisms for overloading operators.... C++ retains C's ability to deal efficiently with the fundamental objects of the hardware (bits, bytes, words, addresses, etc.). This allows the user-defined types to be implemented with a pleasing degree of efficiency."*

1987

- Apple engineer William Atkinson designed HyperCard, a software tool that simplifies development of in-house applications. HyperCard differed from previous programs of its sort because Atkinson made it interactive rather than language-based and geared it toward the construction of user interfaces rather than the processing of data. In HyperCard, programmers built stacks with the concept of hypertext links between stacks of pages. Apple distributed the program free with Macintosh computers until 1992.

Hypercard users could look through existing HyperCard stacks as well as add to or edit the stacks. As a stack author, a programmer employed various tools to create his own stacks, linked together as a sort of slide show. At the lowest level, the program linked cards sequentially in chronological ordered, but the HyperTalk programming language allowed more sophisticated links.

1989



Box Art for SimCity

- Maxis released SimCity, a video game that helped launch of series of simulators. Maxis cofounder Will Wright built on his childhood interest in plastic models of ships and airplanes, eventually starting up a company with Jeff Braun and designing a computer program that allowed the user to create his own city. A number of other Sims followed in the series, including SimEarth, SimAnt, and SimLife.

  In SimCity, a player starts with an untouched earth. As the mayor of a city or city planner, he creates a landscape and then constructs buildings, roads, and waterways. As the city grows, the mayor must provide basic services like health care and education, as well as making decisions about where to direct money and how to build a revenue base. Challenges come in the form of natural disasters, airplane crashes, and monster attacks.

1990

- Microsoft shipped Windows 3.0 on May 22. Compatible with DOS programs, the first successful version of Windows finally offered good enough performance to satisfy PC users. For the new version, Microsoft revamped the interface and created a design that allowed PCs to support large graphical applications for the first time. It also allowed multiple programs to run simultaneously on its Intel 80386 microprocessor.

  Microsoft released Windows amid a $10 million publicity blitz. In addition to making sure consumers knew about the product, Microsoft lined up a number of other applications ahead of time that ran under Windows 3.0, including versions of Microsoft Word and Microsoft Excel. As a result, PCs moved toward the user-friendly concepts of the Macintosh, making IBM and IBM-compatible computers more popular.

1991


Linus Torvalds, 1991

- Designed by Finnish university student Linus Torvalds, Linux was released to several Usenet newsgroups on September 17th, 1991. Almost immediately, enthusiasts began developing and improving Linux, such as adding support for peripherals and improving its stability. In February 1992, Linux became free software or (as its developers preferred to say after 1998) open source. Linux typically incorporated elements of the GNU operating system and became widely used.

- Pretty Good Privacy is introduced. Pretty Good Privacy, or PGP, is an e-mail encryption program. Its inventor, software engineer Phil Zimmermann, created it as a tool for people to protect themselves from intrusive governments around the world. Zimmermann posted PGP on the Internet in 1991 where it was available as a free download. The United States government, concerned about the strength of PGP, which rivaled some of the best secret codes in use at the time, prosecuted Zimmermann but dropped its investigation in 1996. PGP is now the most widely used encryption system for e-mail in the world.

1.4 Pioneers of Computing

1801: In France, Joseph Marie Jacquard invents a loom that uses punched wooden cards to automatically weave fabric designs. Early computers would use similar punch cards.
1822: English mathematician Charles Babbage conceives of a steam-driven calculating machine that would be able to compute tables of numbers. The project, funded by the English government, is a failure. More than a century later, however, the world's first computer was actually built.
1890: Herman Hollerith designs a punch card system to calculate the 1880 census, accomplishing the task in just three years and saving the government $5 million. He establishes a company that would ultimately become IBM.
1936: Alan Turing presents the notion of a universal machine, later called the Turing machine, capable of computing anything that is computable. The central concept of the modern computer was based on his ideas.
1937: J.V. Atanasoff, a professor of physics and mathematics at Iowa State University, attempts to build the first computer without gears, cams, belts or shafts.

1941: Atanasoff and his graduate student, Clifford Berry, design a computer that can solve 29 equations simultaneously. This marks the first time a computer is able to store information on its main memory.
1943-1944: Two University of Pennsylvania professors, John Mauchly and J. Presper Eckert, build the Electronic Numerical Integrator and Calculator (ENIAC**)**. Considered the grandfather of digital computers, it fills a 20-foot by 40-foot room and has 18,000 vacuum tubes.
1946: Mauchly and Presper leave the University of Pennsylvania and receive funding from the Census Bureau to build the UNIVAC, the first commercial computer for business and government applications.
1947: William Shockley, John Bardeen and Walter Brattain of Bell Laboratories invent the transistor. They discovered how to make an electric switch with solid materials and no need for a vacuum.
1953: Grace Hopper develops the first computer language, which eventually becomes known as COBOL. Thomas Johnson Watson Jr., son of IBM CEO Thomas Johnson Watson Sr., conceives the IBM 701 EDPM to help the United Nations keep tabs on Korea during the war.
1954: The FORTRAN programming language is born.
1958: Jack Kilby and Robert Noyce unveil the integrated circuit, known as the computer chip. Kilby was awarded the Nobel Prize in Physics in 2000 for his work.
1964: Douglas Engelbart shows a prototype of the modern computer, with a mouse and a graphical user interface (GUI**)**. This marks the evolution of the computer from a specialized machine for scientists and mathematicians to technology that is more accessible to the general public.
1969: A group of developers at Bell Labs produce UNIX, an operating system that addressed compatibility issues. Written in the C programming language, UNIX was portable across multiple platforms and became the operating system of choice among mainframes at large companies and government entities. Due to the slow nature of the system, it never quite gained traction among home PC users.
1970: The newly formed Intel unveils the Intel 1103, the first Dynamic Access Memory (DRAM) chip.
1971: Alan Shugart leads a team of IBM engineers who invent the "floppy disk," allowing data to be shared among computers.
1973: Robert Metcalfe, a member of the research staff for Xerox, develops Ethernet for connecting multiple computers and other hardware.
1974-1977: A number of personal computers hit the market, including Scelbi & Mark-8 Altair, IBM 5100, RadioShack's TRS-80 —affectionately known as the "Trash 80" — and the Commodore PET.
1975: The January issue of Popular Electronics magazine features the Altair 8080, described as the "world's first minicomputer kit to rival commercial models." Two "computer geeks," Paul Allen and Bill Gates, offer to write software for the Altair, using the new BASIC

language. On April 4, after the success of this first endeavor, the two childhood friends form their own software company, Microsoft.
1976: Steve Jobs and Steve Wozniak start Apple Computers on April Fool's Day and roll out the Apple I, the first computer with a single-circuit board.

The TRS-80, introduced in 1977, was one of the first machines whose documentation was intended for non-geeks

1977: Radio Shack's initial production run of the TRS-80 was just 3,000. It sold like crazy. For the first time, non-geeks could write programs and make a computer do what they wished.
1977: Jobs and Wozniak incorporate Apple and show the Apple II at the first West Coast Computer Faire. It offers color graphics and incorporates an audio cassette drive for storage.
1978: Accountants rejoice at the introduction of VisiCalc, the first computerized spreadsheet program.
1979: Word processing becomes a reality as MicroPro International releases WordStar.

The first IBM personal computer, introduced on Aug. 12, 1981, used the MS-DOS operating system.
Credit: IBM

1981: The first IBM personal computer, code-named "Acorn," is introduced. It uses Microsoft's MS-DOS operating system. It has an Intel chip, two floppy disks and an optional color monitor. Sears & Roebuck and Computerland sell the machines, marking the first time a computer is available through outside distributors. It also popularizes the term PC.
1983: Apple's Lisa is the first personal computer with a GUI. It also features a drop-down menu and icons. It flops but eventually evolves into the Macintosh. The Gavilan SC is the first portable computer with the familiar flip form factor and the first to be marketed as a "laptop."
1985: Microsoft announces Windows, its response to Apple's GUI. Commodore unveils the Amiga 1000, which features advanced audio and video capabilities.
1985: The first dot-com domain name is registered on March 15, years before the World Wide Web would mark the formal beginning ofInternet history. The Symbolics Computer Company, a small Massachusetts computer manufacturer, registers Symbolics.com. More than two years later, only 100 dot-coms had been registered.

1986: Compaq brings the Deskpro 386 to market. Its 32-bit architecture provides as speed comparable to mainframes.
1990: Tim Berners-Lee, a researcher at CERN, the high-energy physics laboratory in Geneva, develops HyperText Markup Language (HTML), giving rise to the World Wide Web.
1993: The Pentium microprocessor advances the use of graphics and music on PCs.
1994: PCs become gaming machines as "Command & Conquer," "Alone in the Dark 2," "Theme Park," "Magic Carpet," "Descent" and "Little Big Adventure" are among the games to hit the market.
1996: Sergey Brin and Larry Page develop the Google search engine at Stanford University.
1997: Microsoft invests $150 million in Apple, which was struggling at the time, ending Apple's court case against Microsoft in which it alleged that Microsoft copied the "look and feel" of its operating system.
1999: The term Wi-Fi becomes part of the computing language and users begin connecting to the Internet without wires.
2001: Apple unveils the Mac OS X operating system, which provides protected memory architecture and pre-emptive multi-tasking, among other benefits. Not to be outdone, Microsoft rolls out Windows XP, which has a significantly redesigned GUI.
2003: The first 64-bit processor, AMD's Athlon 64, becomes available to the consumer market.
2004: Mozilla's Firefox 1.0 challenges Microsoft's Internet Explorer, the dominant Web browsers. Facebook, a social networking site, launches.
2005: YouTube, a video sharing service, is founded. Google acquires Android, a Linux-based mobile phone operating system.
2006: Apple introduces the MacBook Pro, its first Intel-based, dual-core mobile computer, as well as an Intel-based iMac. Nintendo's Wii game console hits the market.
2007: The iPhone brings many computer functions to the smartphone.
2009: Microsoft launches Windows 7, which offers the ability to pin applications to the taskbar and advances in touch and handwriting recognition, among other features.
2010: Apple unveils the iPad, changing the way consumers view media and jumpstarting the dormant tablet computer segment.
2011: Google releases the Chromebook, a laptop that runs the Google Chrome OS.
2012: Facebook gains 1 billion users on October 4.
2015: Apple releases the Apple Watch. Microsoft releases Windows 10.